Vinayak Kamat

AI/ML Engineer | Generative AI, RAG & Production AI Systems

Mumbai, India vinayakkamat001@gmail.com 072081 14510 vinayak-kamat-81b8901a0

vinayakkamatcodes

Summary

AI/ML Engineer with hands-on experience developing production-oriented AI systems, Generative AI applications, Retrieval-Augmented Generation (RAG) pipelines, and computer vision models. Experienced in Python, PyTorch, FastAPI, LLM integration, embeddings, prompt engineering, model evaluation, and cloud-based ML deployment using AWS. Built an LLM hallucination detection and risk analysis gateway, optimized model inference using ONNX, and implemented offline LLM inference on Android devices. Focused on reliable AI systems, reusable software components, model performance, and continuous optimization.

Skills

Programming Languages: Python, C++, Dart, JavaScript

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings, Prompt Engineering, LangChain, LlamaIndex, GGUF Models, llama.cpp

AI Evaluation & Reliability: RAG Pipeline Evaluation, Model Calibration, Hallucination Detection, Semantic Risk Assessment, Model Performance Optimization

Machine Learning & Computer Vision: PyTorch, Transformers, OpenCV, spaCy, ONNX, Ulcer Detection, Model Inference Optimization

Backend & API Engineering: FastAPI, Flask, REST APIs, Python Components, API Integration, Streaming Responses

Cloud & Deployment: AWS Lambda, Amazon S3, Amazon ECR, Amazon DynamoDB, Docker, Production ML Pipelines

MLOps & Monitoring: MLflow, Experiment Tracking, Model Versioning, Reproducible Workflows, Datadog

Data Processing: Pandas, NumPy

Experience

KAN Innovations, Junior AI/ML Engineer — Mumbai, Maharashtra
May 2025 – present
1 year 6 months

- Developed and maintained production-oriented AI/ML components, integrating machine learning models into application and deployment pipelines.
- Improved the calibration and reliability of machine learning predictions in production.
- Optimized model inference by converting PyTorch models to ONNX for improved runtime performance.
- Integrated ML models into cloud-based workflows using AWS Lambda, Amazon ECR, Amazon S3, and Amazon DynamoDB.
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines for LLM-powered applications.
- Used MLflow for experiment tracking, model versioning, and reproducible machine learning workflows.
- Developed reusable Python components for AI/ML integration and model deployment.

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Developed a PyTorch-based ulcer detection model achieving approximately 85% accuracy.
- Improved the performance of a computer vision model for eye posture detection.
- Contributed to machine learning model development, evaluation, and performance improvement.

Projects

Aethelgard
AI governance gateway for real-time LLM output analysis, hallucination detection, and semantic risk assessment.

- Built a real-time LLM response analysis pipeline using a secondary model for semantic risk assessment.
- Implemented a FastAPI gateway to process LLM outputs and stream risk metrics to Datadog.
- Designed automated hallucination detection logic to support AI output validation and more reliable AI deployments.
- Developed an AI safety-oriented architecture focused on identifying potentially unreliable or risky model responses.

nanoMind
Offline AI chat application enabling on-device LLM inference on Android devices.

- Integrated GGUF language models using llama.cpp to enable offline LLM inference without cloud dependency.
- Implemented real-time token streaming using Kotlin Coroutines and Jetpack Compose.
- Optimized LLM loading and inference for Android devices using ARM-specific performance improvements.

Financial AI Agent (Demo)
Prototype AI agent for financial data retrieval, query routing, and language-model-based reasoning.

- Built retrieval and reasoning pipelines for financial information and insights.
- Implemented modular Python components for query routing and response refinement.
- Explored LLM-powered workflows for retrieving relevant information and generating contextual responses.

Education

MSc **WorldQuant University**, Financial Engineering — Remote
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

Achievements

Participated in Kaggle competitions focused on applied machine learning.

Maintained open-source AI/ML projects on GitHub.